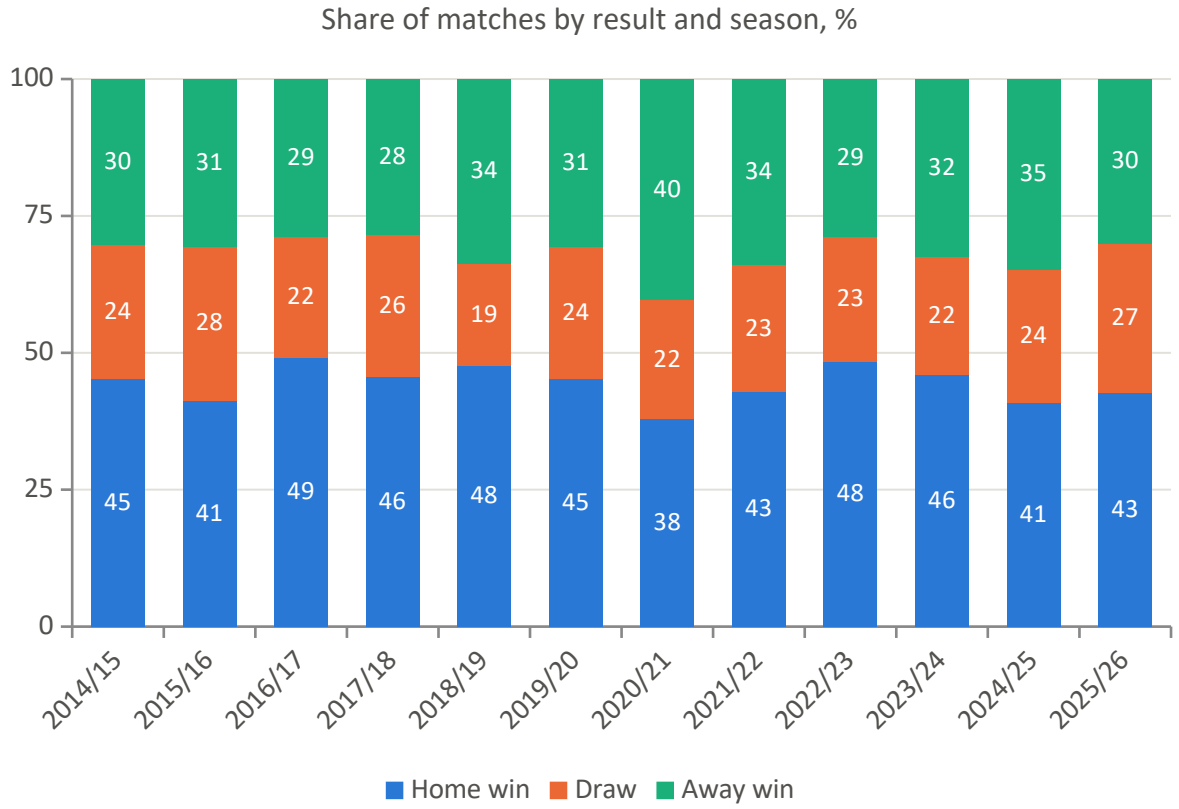
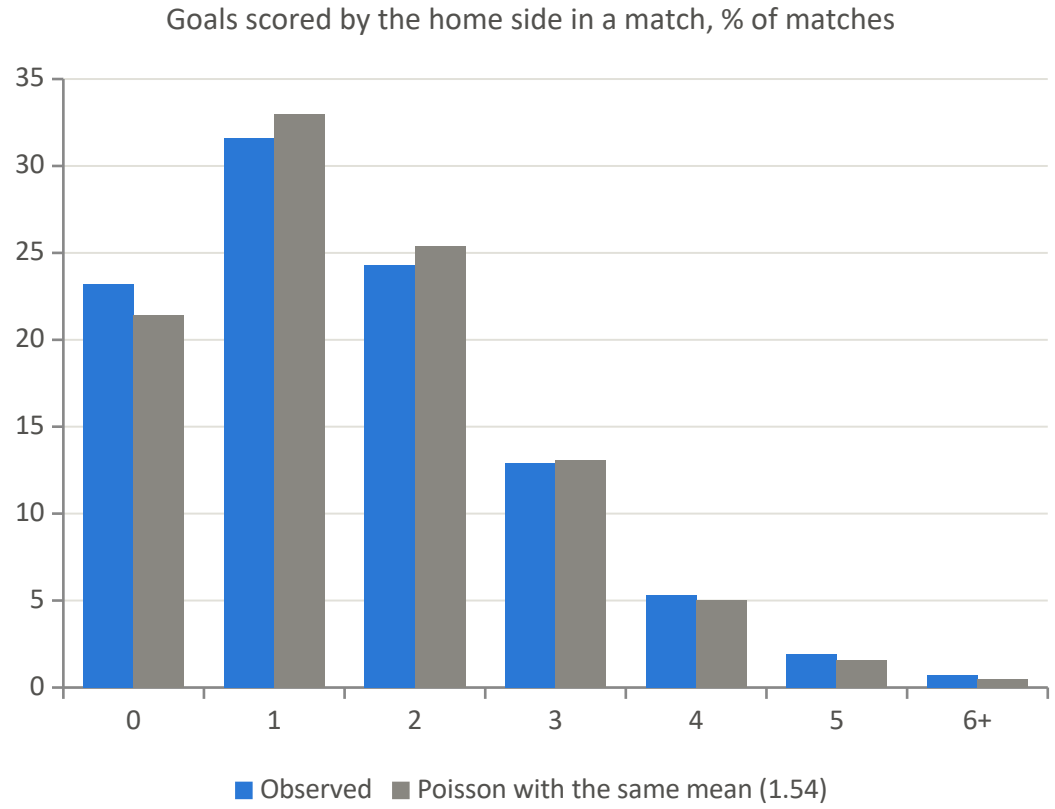
# Predicting Premier League match results

Project phase 2: exploratory analysis, models, error analysis

# Home sides win more; goals follow Poisson

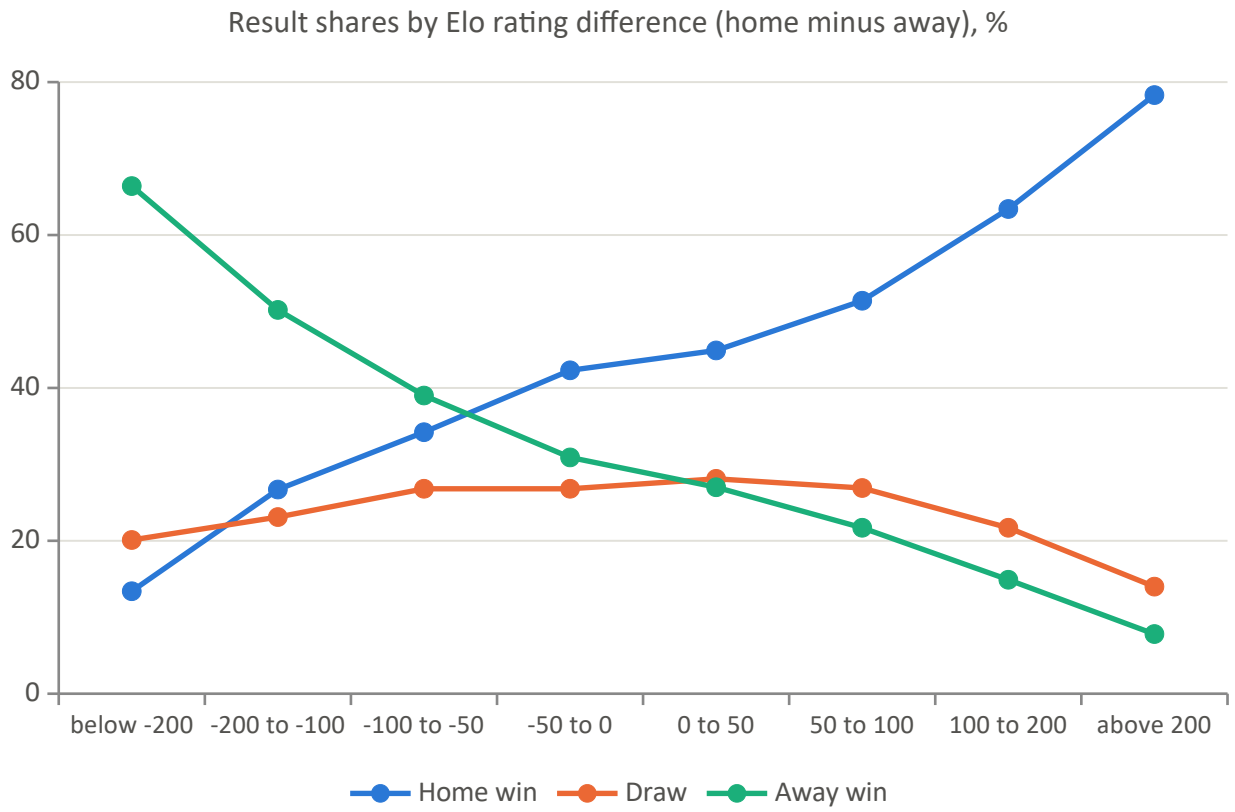



Over 4,560 matches: home 44%, draw 24%, away 32%. Lowest home share: 38% in 2020/21, played mostly without crowds.

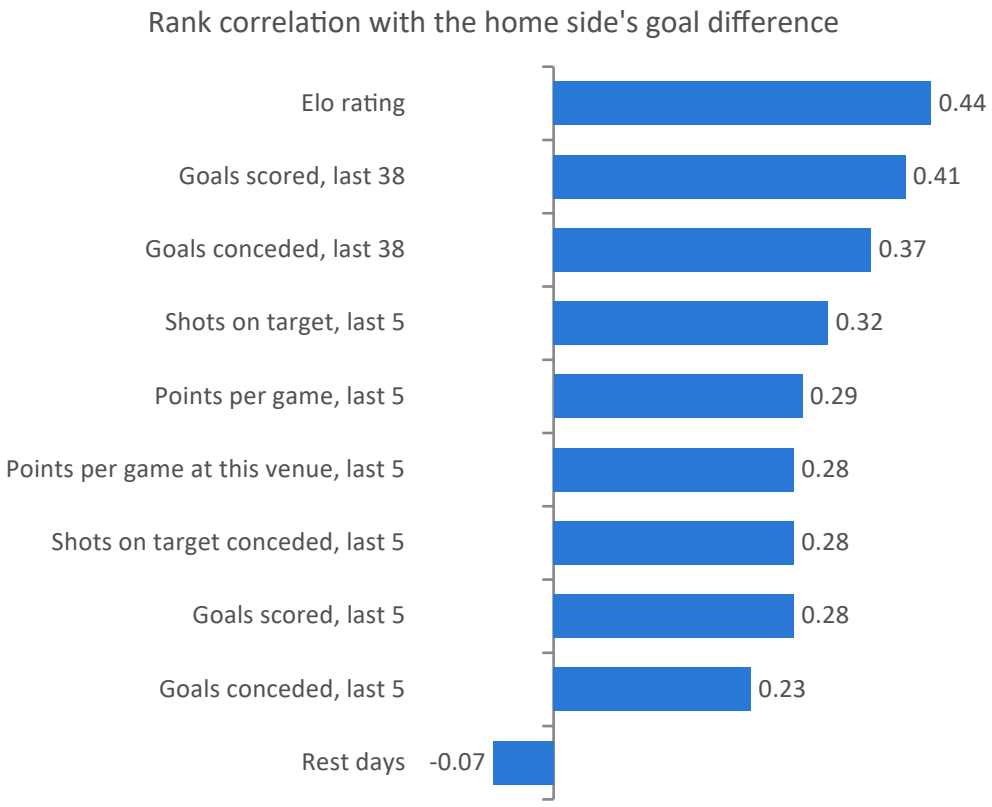



Every observed share is within 1.9 points of the Poisson value. The goal model rests on this.

# Long-run strength predicts results; form adds little




With a rating lead above 200 the home side wins 78% of matches. Draws never exceed 28% in any group.




Each bar is the home side's advantage on one feature. Ratings and season-long goals lead; five-match form trails; rest days carry nothing.

# Six models, one walk-forward protocol

### Class frequencies

The same three probabilities for every match. The floor.

### Elo + logistic regression

One feature: the rating difference.

### Poisson goal model

Attack and defence strength per club give expected goals and a score grid.

### Logistic regression

All 21 features, multinomial, regularised.

### Random forest

400 trees on the 21 features.

### Gradient boosting

Shallow boosted trees on the 21 features.

## Twelve seasons, used in order

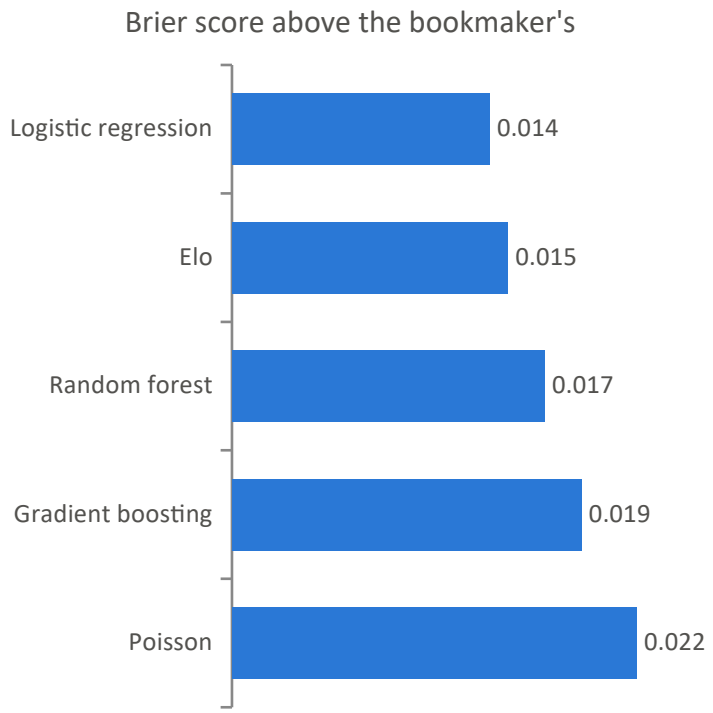


- **Warm-up** 2014/15
- **Training** 2015/16 to 2020/21
- **Validation** 2021/22, 2022/23
- **Test** 2023/24 to 2025/26

- Settings chosen on the validation seasons, then frozen
- Before each test season a model is refitted on every earlier season; the Poisson model before every match day
- 1,140 test matches, scored by accuracy, Brier score and log loss
- Reference: Bet365's odds at kickoff, margin removed

# No model reaches the bookmaker

| Model | Accuracy | Brier | Log loss | Brier gap to bookmaker, 95% interval |
|---|---|---|---|---|
| **Logistic regression** | **53.9%** | **0.584** | **0.980** | **+0.014 (+0.007 to +0.021)** |
| Elo + logistic regression | 53.4% | 0.585 | 0.981 | +0.015 (+0.008 to +0.022) |
| Random forest | 53.6% | 0.587 | 0.986 | +0.017 (+0.010 to +0.023) |
| Gradient boosting | 52.5% | 0.589 | 0.990 | +0.019 (+0.011 to +0.026) |
| Poisson goal model | 52.0% | 0.592 | 0.994 | +0.022 (+0.013 to +0.031) |
| Class frequencies | 43.2% | 0.650 | 1.073 | +0.080 (+0.063 to +0.095) |
| **Bookmaker (Bet365 at kickoff)** | **55.0%** | **0.570** | **0.960** | **reference** |




- Best model: logistic regression, Brier 0.584 against the bookmaker's 0.570; every interval excludes zero
- The rating difference alone (0.585) matches all 21 features; tree models are slightly worse
- All models are far better than class frequencies (0.650)

# Draws are a quarter of results and never predicted

## Logistic regression

| Actual | Pred. home | Pred. draw | Pred. away |
|---|---|---|---|
| **Home** | 410<br>83% | 0<br>0% | 82<br>17% |
| **Draw** | 178<br>64% | 0<br>0% | 101<br>36% |
| **Away** | 164<br>44% | 0<br>0% | 205<br>56% |

## Bookmaker

| Actual | Pred. home | Pred. draw | Pred. away |
|---|---|---|---|
| **Home** | 403<br>82% | 0<br>0% | 89<br>18% |
| **Draw** | 171<br>61% | 0<br>0% | 108<br>39% |
| **Away** | 145<br>39% | 0<br>0% | 224<br>61% |

Rows: what happened. Columns: the result each forecaster rated most likely. Cells: matches, and share of the row.

**0 of 279**

draws predicted by the model, and by the bookmaker

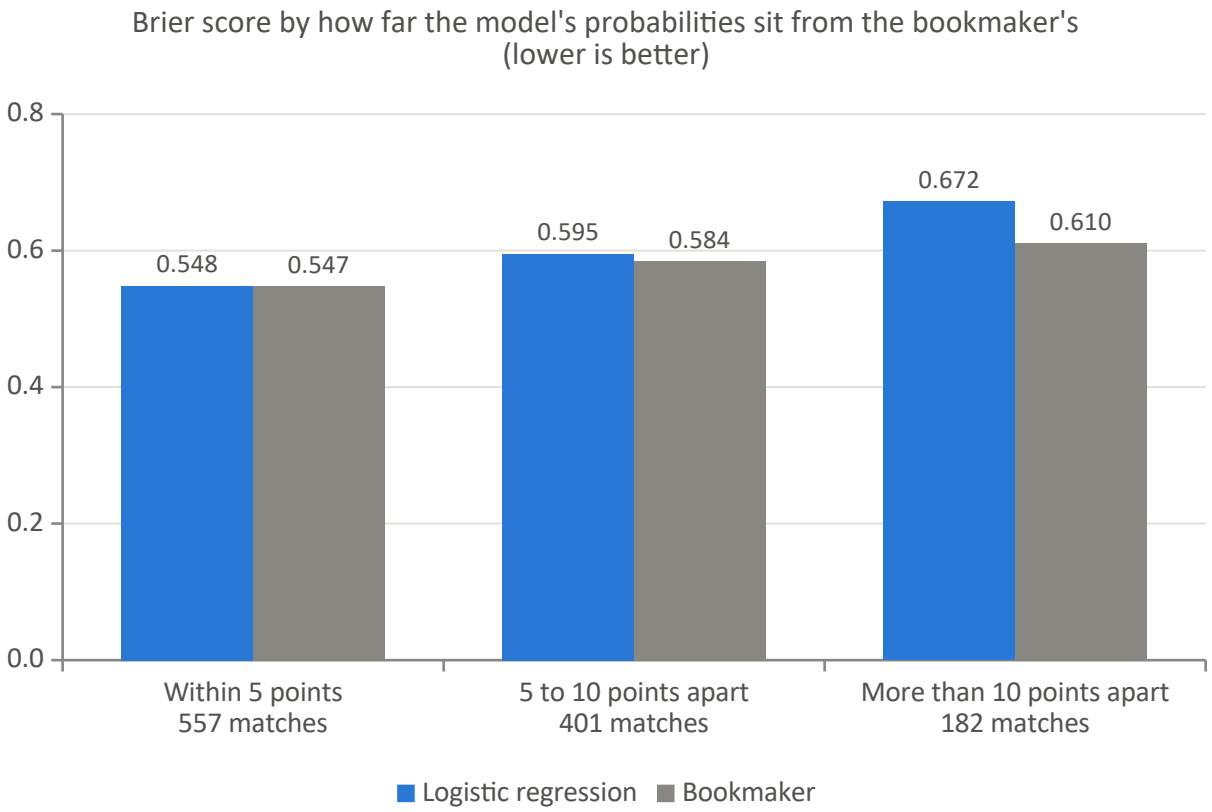
**30%**

is the highest draw probability the model ever gives

**2.5 pts**

largest gap between predicted probability and observed frequency

# When they disagree, the bookmaker is right




In 557 matches the two agree within 5 points and score the same. In the 182 where they differ by more than 10, the bookmaker is ahead by 0.062.

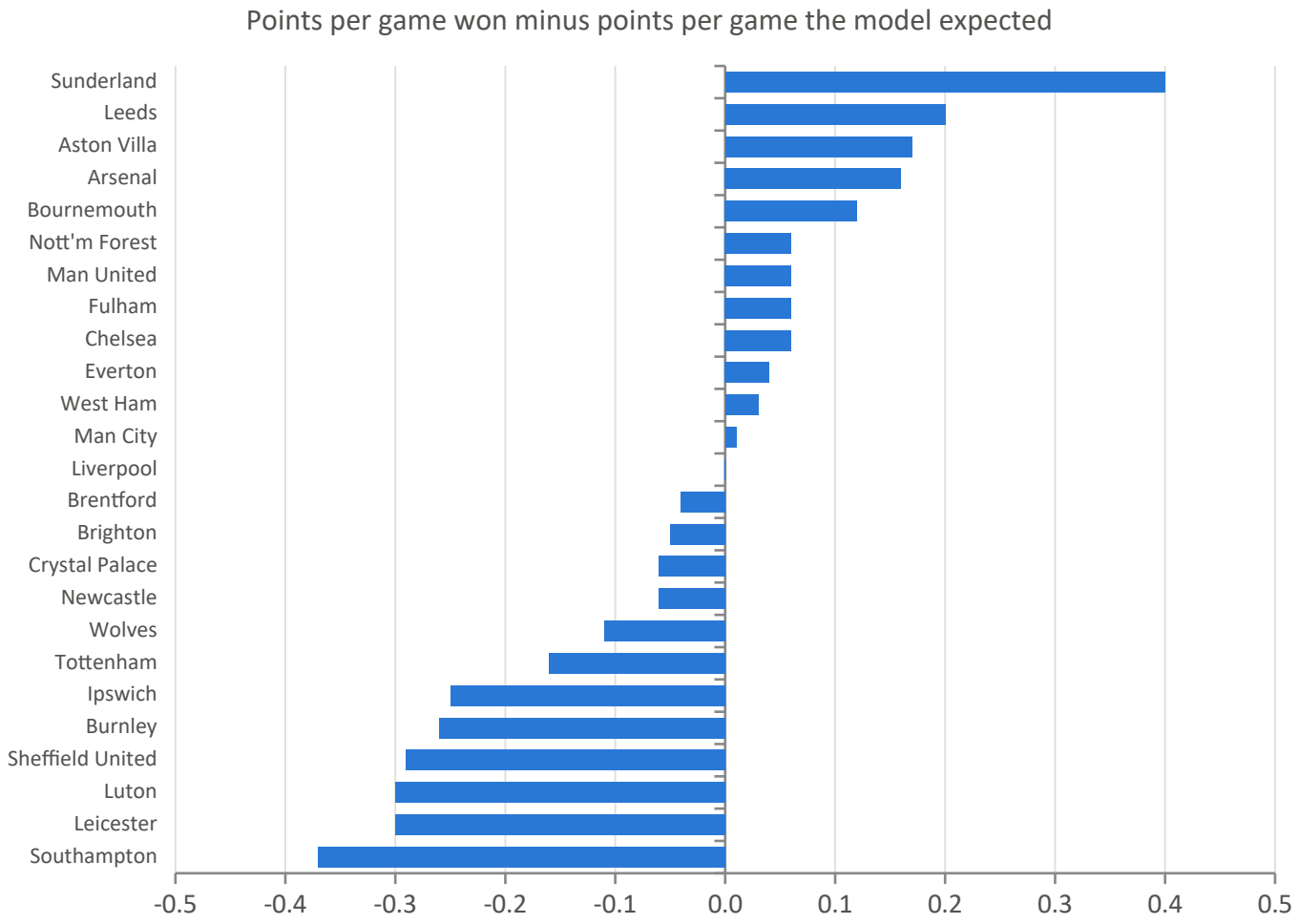
## Betting on the disagreement

One unit on every result a model rates above +5% expected value at the kickoff odds.

| Model | Bets | Won | Return |
|---|---|---|---|
| Elo | 724 | 30% | **−7.0%** |
| Logistic regression | 740 | 27% | **−9.0%** |
| Gradient boosting | 811 | 26% | **−15.1%** |
| Random forest | 767 | 25% | **−17.9%** |
| Poisson | 872 | 26% | **−18.5%** |

Every model loses money. A disagreement with a stronger forecaster is mostly the weaker one's error.

# Clubs new to the league are the blind spot




## 8 largest errors

are all clubs that spent only one or two of the three seasons in the league. A promoted club arrives with no history, so the model gives it an average newcomer's rating.

## Upsets surprise everyone

The result the model thought least likely: Liverpool 0-1 Crystal Palace, 6% for the model and 9% for the bookmaker. That part of the error is chance.

# What we learned

### A simple, calibrated model gets most of the way

Logistic regression: 54% accuracy, Brier 0.584 against 0.650 for class frequencies.

### The bookmaker stays ahead

Brier 0.570. Its odds reflect line-ups, injuries and transfers that results alone do not show.

### More complexity did not help

One rating feature matches 21 features, and tree ensembles are slightly worse on a few thousand training matches.

### Next steps

Squad and line-up data, a model aimed at draws, and better starting ratings for promoted clubs.